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ROLL NO : 23I-0058

SECTION : BSE – 6A

COMPUTER NETWORKS LAB

BSSE SPRING 2026

LAB 02

1.

Task 1: Network Hardware Exploration (10 Marks)

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 776, y: 61

Root

05:01:00

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BSE-5A
LAB 02

PC-PT
PC1

Represent end users computers in network

Laptop-PT
Laptop

Represent a portable device

2960-24TT
Switch

Used for connecting devices in same network

WRT300N
Wireless Router

Connecting wireless devices to local network

1841
Router

Used for connecting devices in different networks

Hub-PT
Hub

Used for broadcasting data to all devices at same time

PC-PT
PC2

represent end-user computers in the network

Time: 00:57:57

Realtime Simulation

Scenario 0

New Delete

ngle PDU List Wind

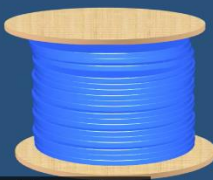
WRT300N

youtube.com/watch?v=_NX99ad2FUA

Search

Twisted Pair CABLE

POWERCERT VIDEOS



buy a bulk roll of twisted pair cable and then you would have to wire them correctly and then

Ethernet Cables, UTP vs STP, Straight vs Crossover, CAT 5,5e,6,7,8 Network Cables

PowerCert Animated Videos 1.73M subscribers

Subscribe 45K

Share Ask

Networking Tools - Hardware 511K views · 5 years ago

*Untitled - Notepad

File Edit Format View Help

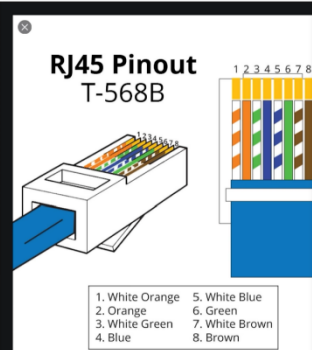
Name Shah Faisal
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 Section BSE-5A
 Computer Network Lab 02

Ln 4, Col 24 100% Windows (CRLF) UTF-8

youtube.com/watch?v=NWhoJp8UQpo

Search

RJ45 Pinout T-568B



at home and terminating both ends with an RJ45 plug then it really doesn't make any

How to Wire Up Ethernet Plugs the EASY WAY! (Cat5e / Cat6 RJ45 Pass Through Connectors)

Switched On Network 72.2K subscribers

Subscribe 73K

Share Ask

Stop Wiring RJ45s WRONG! (EZ vs. Pro Method)

*Untitled - Notepad

File Edit Format View Help

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Ln 4, Col 24 100% Windows (CRLF) UTF-8

1.What is the function of an RJ45 connector?

An RJ45 connector terminates Ethernet cables and provides a standardized physical interface to connect devices to a network.

2.What is the primary difference between the T568A and T568B wiring standards?

They differ only in the order of the green and orange wire pairs; T568B swaps these pairs compared to T568A.

3.How does the wiring of a "Crossover" cable differ from a "Straight-through" cable?

A straight-through cable uses the same wiring standard on both ends, while a crossover cable uses T568A on one end and T568B on the other.

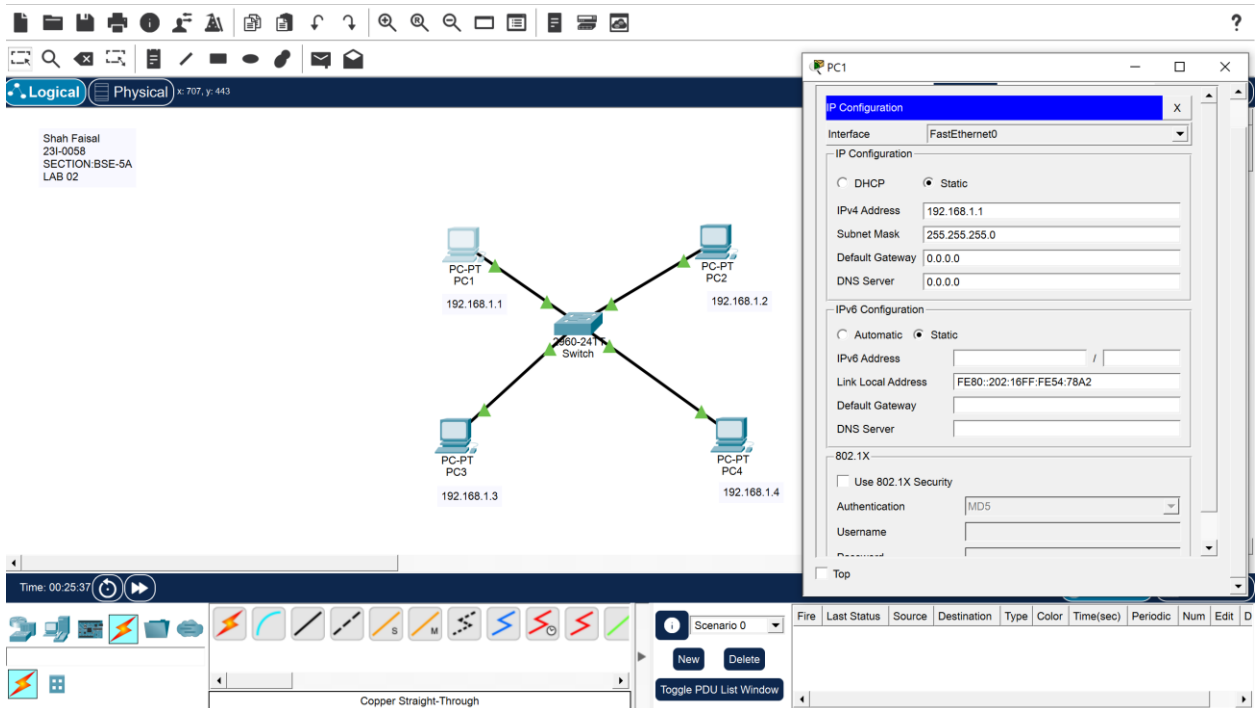
4.When connecting a computer directly to another computer (without a switch), which cable type should be used?

A crossover cable is used when connecting two computers directly without a switch or hub.

5.List the correct color order for the T568B standard (from pin 1 to 8).

White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown

Task 2: Create and Simulate Different Network Topologies (10 Marks)



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LAB 02

Time: 00:25:37

Scenario 0

New Delete

Toggle PDU List Window

Copper Straight-Through

PC1 IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::202:16FF:FE54:78A2

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

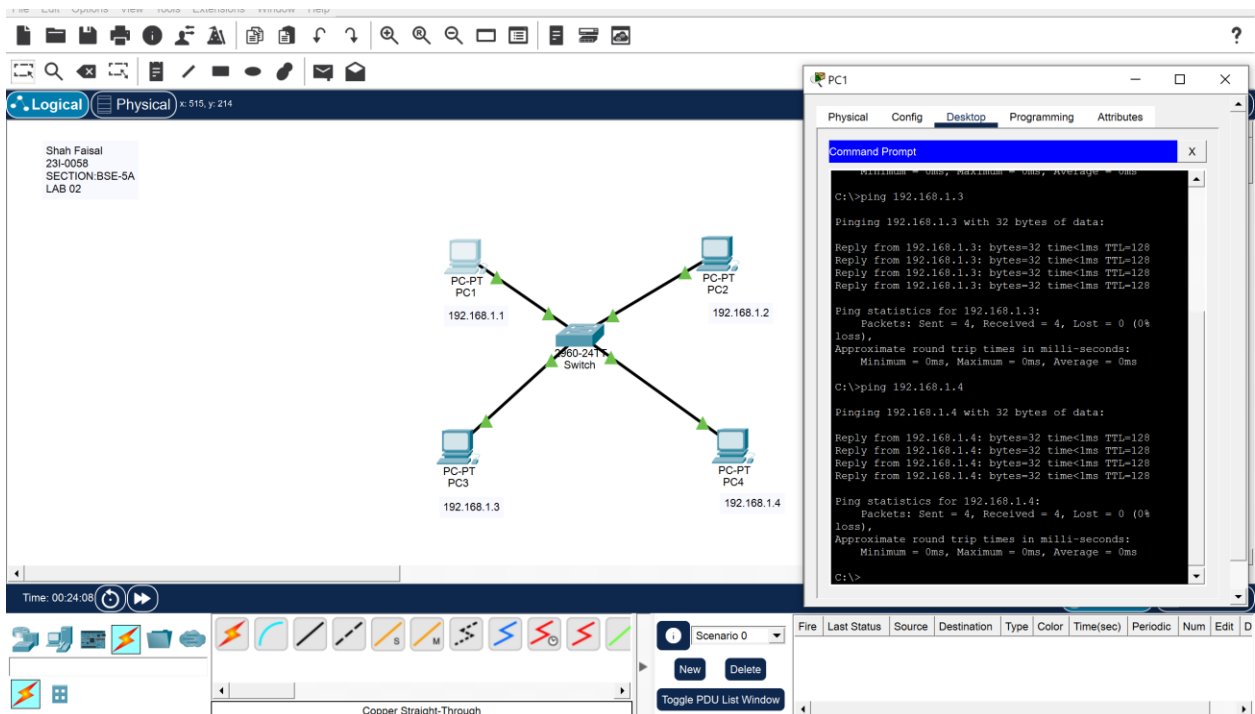
Authentication: MD5

Username:

Password:

Top

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
------	-------------	--------	-------------	------	-------	-----------	----------	-----	------



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LAB 02

Time: 00:24:08

Scenario 0

New Delete

Toggle PDU List Window

Copper Straight-Through

PC1 Command Prompt

```

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

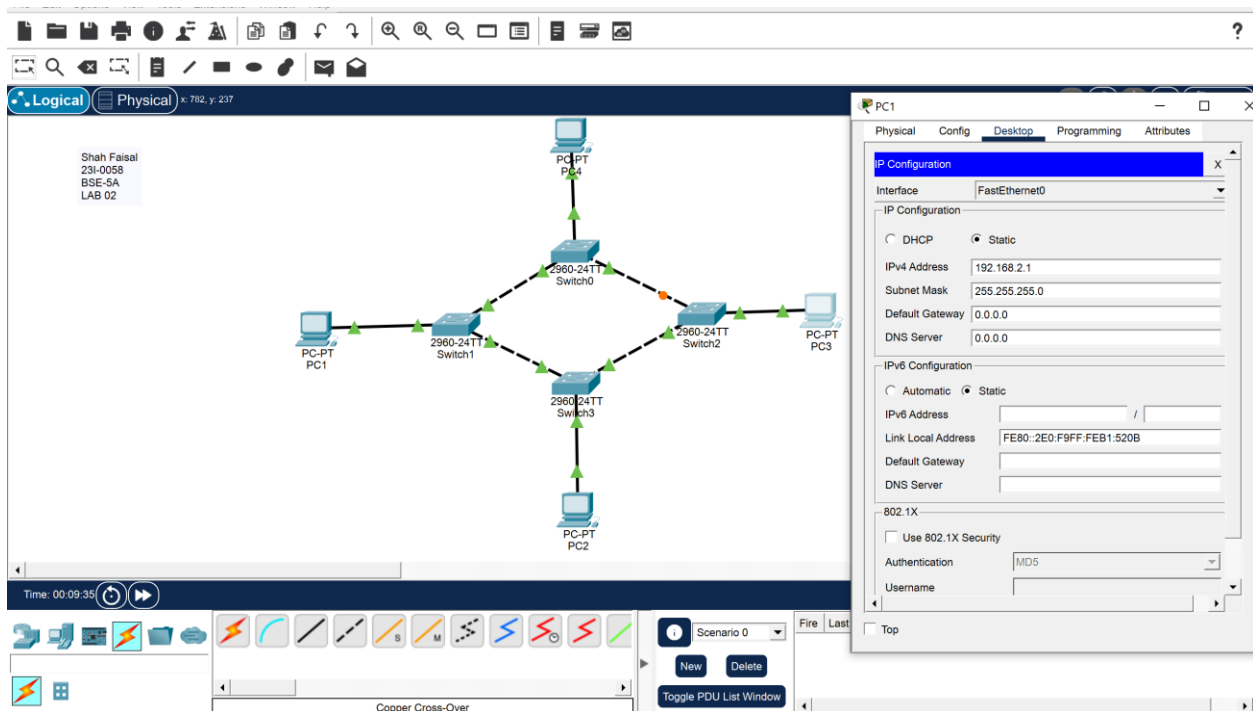
C:\>
  
```

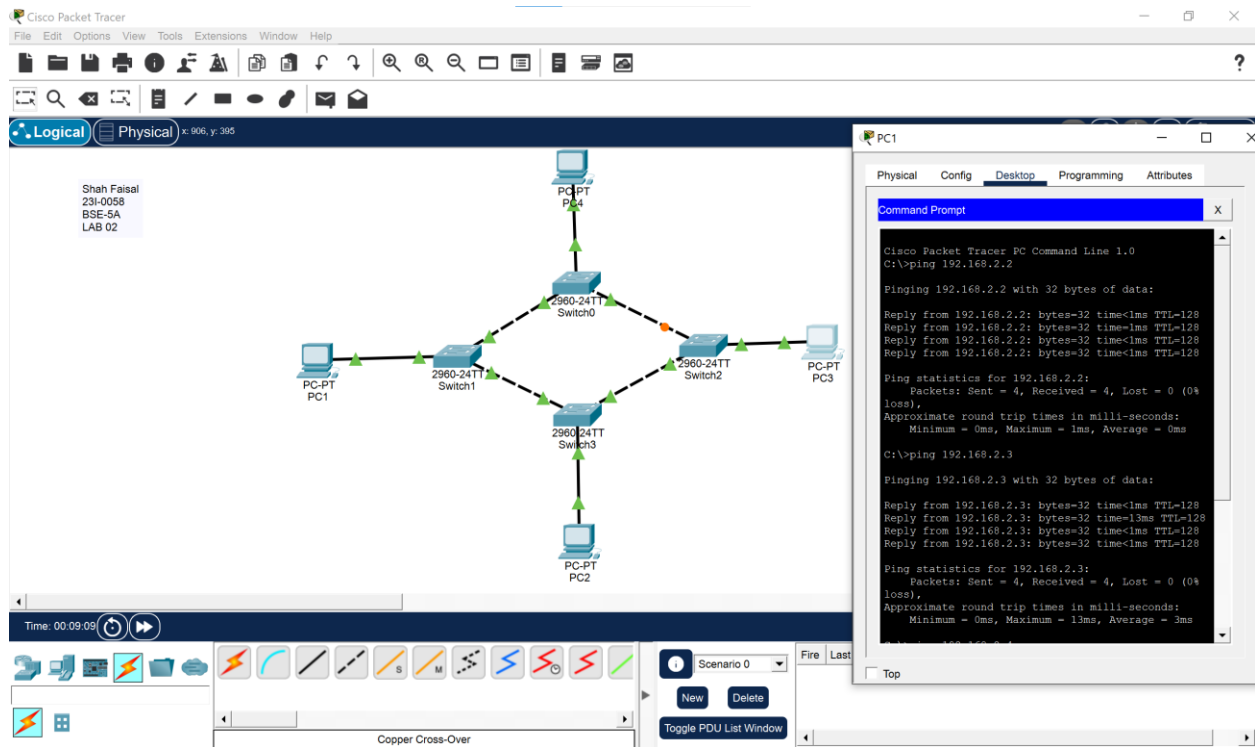
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
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Disadvantage:

The central hub itself becomes a single point of failure; if it fails, the entire network is disrupted.

RING TOPOLOGY

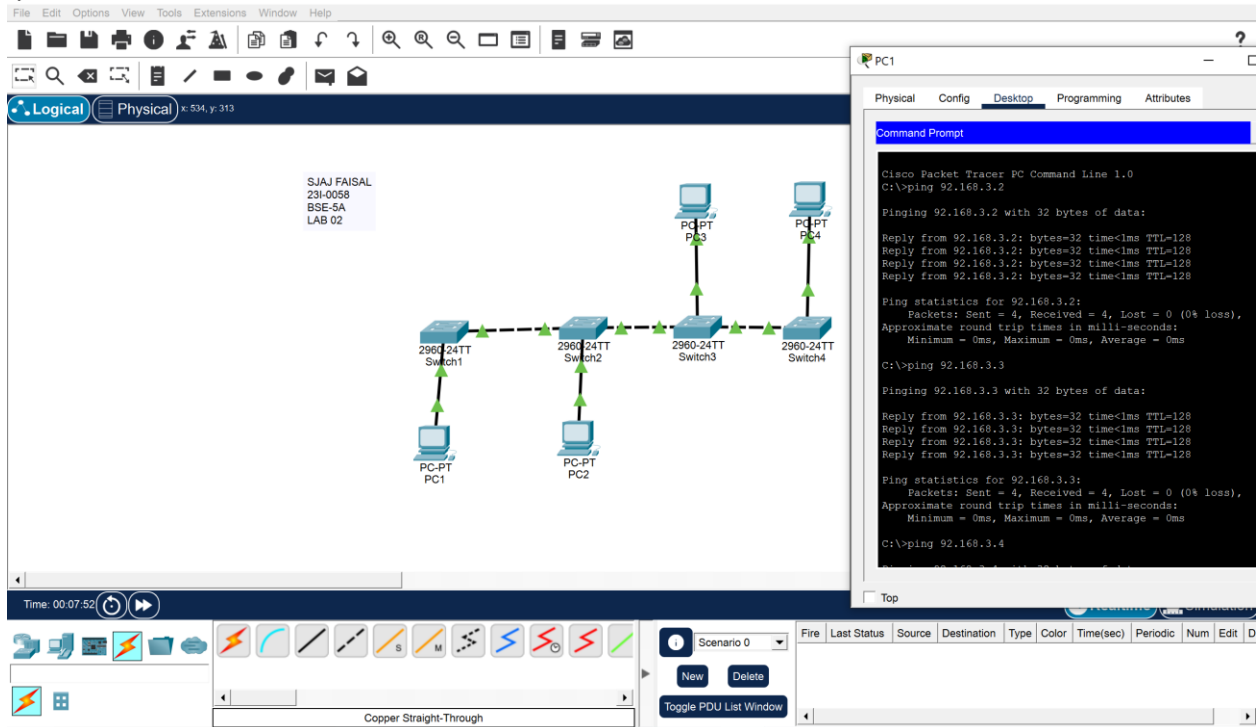




ADVANTAGE:

This topology is relatively simple and prevents data collisions.

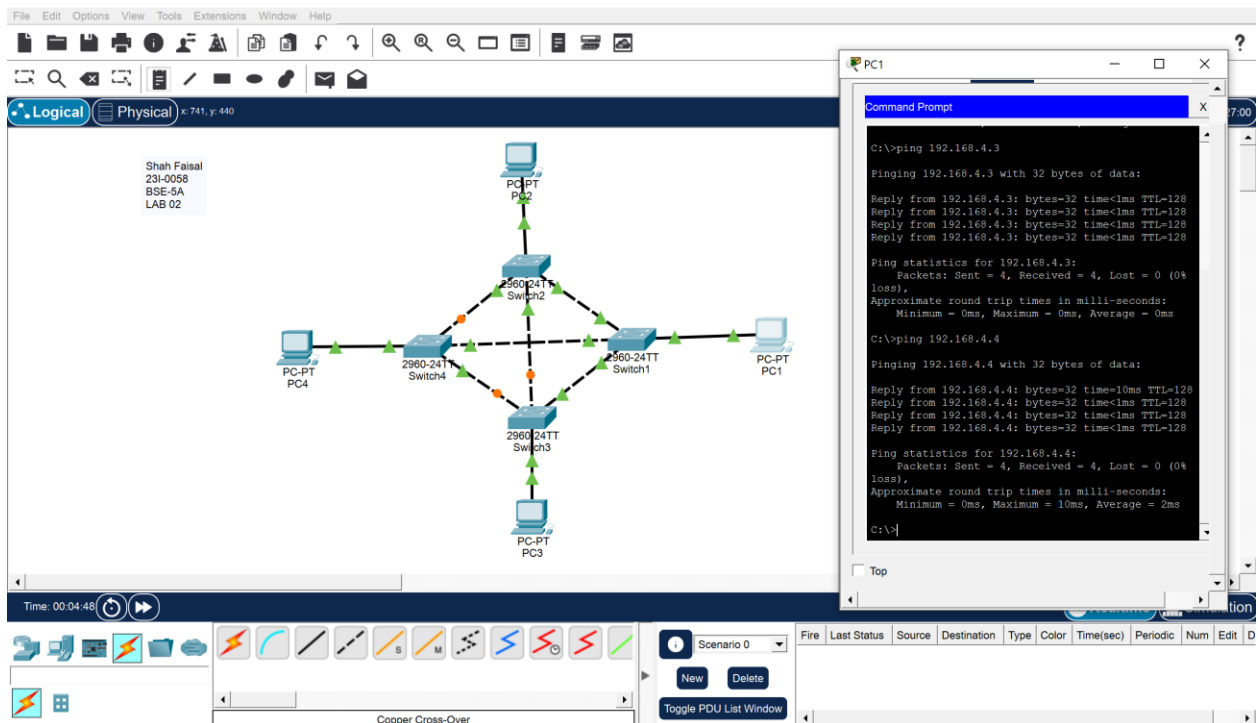
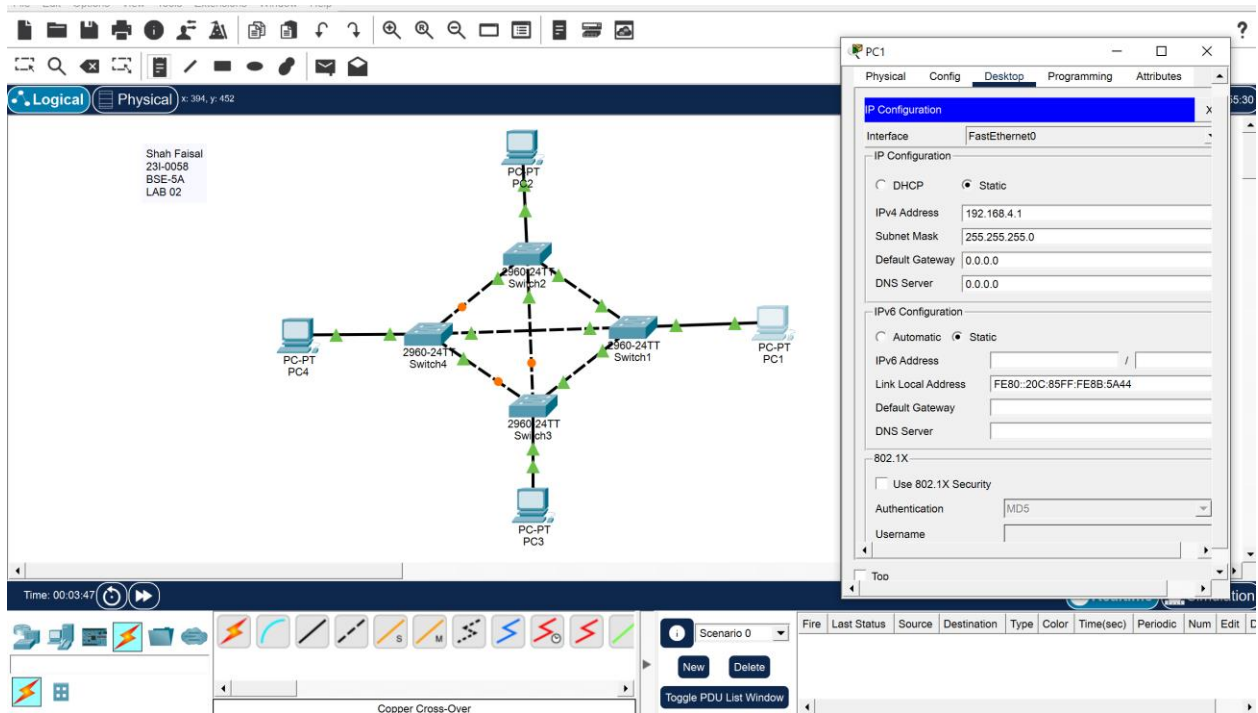
BUS TOPOLOGY



ADVANTAGE:

This topology is cost-effective and simple to implement but is best suited for small networks with limited devices, as performance deteriorates with increased traffic or devices.

MESH TOPLOLOGY:

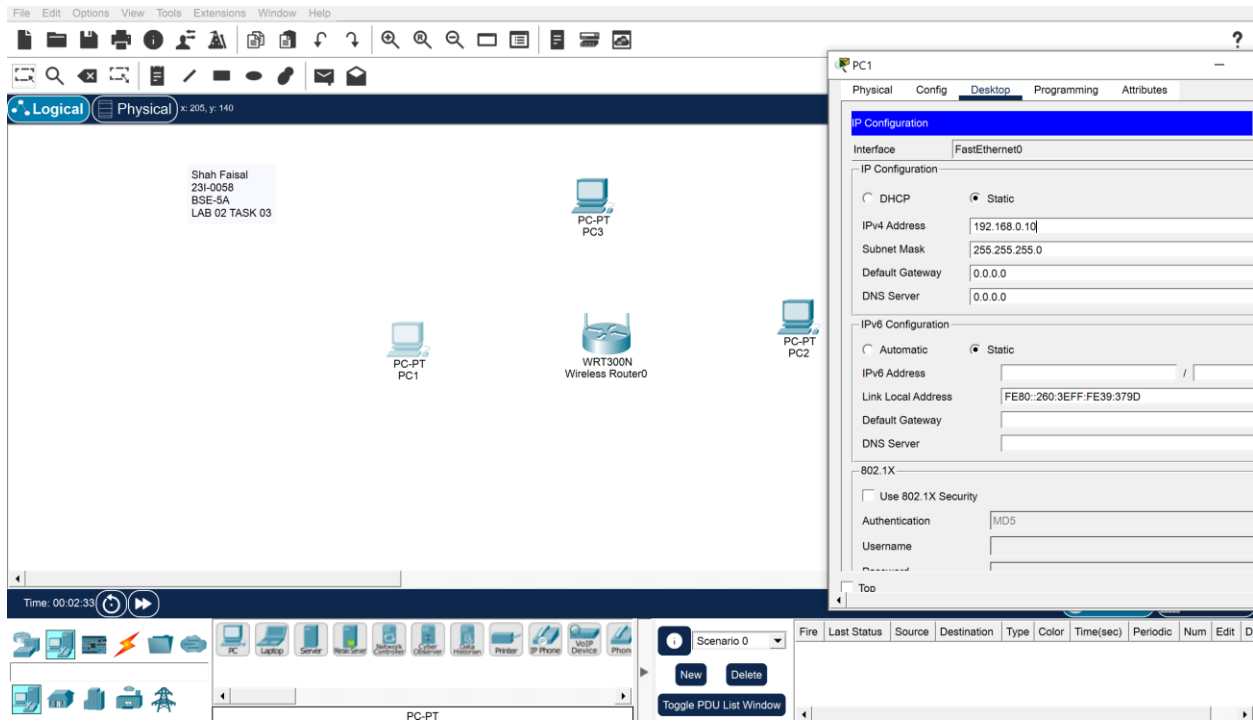


ADVANTAGE:

This ensures high reliability and fault tolerance, as the network

remains operational even if one or more links fail.

Task 3: Configure a Wireless Network with 3 PCs (5 Marks)



The screenshot displays the Packet Tracer interface for configuring a wireless network. The main workspace shows three PCs (PC-PT PC1, PC-PT PC2, PC-PT PC3) and a WRT300N Wireless Router0. The right-hand pane shows the configuration for PC1, specifically the 'Desktop' tab. Under 'P Configuration', the 'FastEthernet0' interface is selected. The 'IP Configuration' section shows 'Static' IP settings: IPv4 Address 192.168.0.10, Subnet Mask 255.255.255.0, Default Gateway 0.0.0.0, and DNS Server 0.0.0.0. The 'IPv6 Configuration' section shows 'Static' settings: IPv6 Address, Link Local Address FE80::260:3EFF:FE39:379D, Default Gateway, and DNS Server. The '802.1X' section shows 'Use 802.1X Security' unchecked, Authentication set to MD5, and Username field empty. The bottom status bar shows 'Time: 00:02:33' and a 'Scenario 0' dropdown menu.

Wireless Router0

Physical Config **GUI** Attributes

Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status

Basic Wireless Settings
Wireless Security
Guest Network
Wireless MAC Filter
Advanced Wireless Settings

Wireless Security

Security Mode: WPA2 Personal
Encryption: AES
Passphrase: ii230058
Key Renewal: 3600 seconds

Help...

Top

Wireless Router0

Physical Config **GUI** Attributes

Internet Service Providers

Network Setup

Router IP

DHCP Server Settings

MTU: Size: 1500

IP Address: 192 . 168 . 0 . 1
Subnet Mask: 255.255.255.252

DHCP Server: ☐ Enabled ☒ Disabled
DHCP Reservation

Start IP Address: 192.168.0. 100
Maximum number of Users: 50
IP Address Range: 192.168.0. 100 - 149
Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 0 . 0 . 0 . 0
Static DNS 2: 0 . 0 . 0 . 0
Static DNS 3: 0 . 0 . 0 . 0
WINS: 0 . 0 . 0 . 0

Top

